COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 2 : SLO 1 — Half Adder and Full Adder
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1. Label the Components (Remembering)

• Input 1: A Input 2: B
• Output 1: Sum Output 2: Carry
• Gate 1: XOR gate Gate 2: AND gate

2. Complete the Truth Table — Full Adder (Understanding)

• A=0, B=0, Cin=0 → S=0, Cout=0
• A=0, B=0, Cin=1 → S=1, Cout=0
• A=0, B=1, Cin=0 → S=1, Cout=0
• A=0, B=1, Cin=1 → S=0, Cout=1
• A=1, B=0, Cin=0 → S=1, Cout=0
• A=1, B=0, Cin=1 → S=0, Cout=1
• A=1, B=1, Cin=0 → S=0, Cout=1
• A=1, B=1, Cin=1 → S=1, Cout=1

3. Trace the Circuit (Applying): A = 1, B = 0, Cin = 1

• Output of first XOR gate: 1 (1 XOR 0 = 1)
• Output of first AND gate: 0 (1 AND 0 = 0)
• Final S (Sum) output: 0 (1 XOR 1 = 0)
• Output of second AND gate: 1 (1 AND 1 = 1)
• Final Cout (Carry Out) output: 1 (1 OR 0 = 1)

4. Compare and Contrast (Analyzing)

• Number of inputs — Half Adder: 2 | Full Adder: 3
• Names of inputs — Half Adder: A, B | Full Adder: A, B, Cin (carry in)
• Number of outputs — Half Adder: 2 | Full Adder: 2
• Names of outputs — Half Adder: Sum, Carry | Full Adder: Sum, Cout (carry out)
• Accepts carry from previous stage — Half Adder: No | Full Adder: Yes
• Primary purpose — Half Adder: adds two single bits | Full Adder: adds three single bits including a carry-in

5. Design a Multi-Bit Adder (Creating) — 2-bit ripple-carry adder

• Stage 1 (LSB): one HALF ADDER with inputs A0 and B0 → output S0 = A0 XOR B0, carry C1 = A0 AND B0.
• Stage 2 (MSB): one FULL ADDER with inputs A1, B1 and Cin = C1 → output S1 and final Cout.
• Connections: A0, B0 → HA (S0, C1); C1, A1, B1 → FA (S1, Cout).
• Outputs: S0 (bit 0), S1 (bit 1), Cout (final carry).